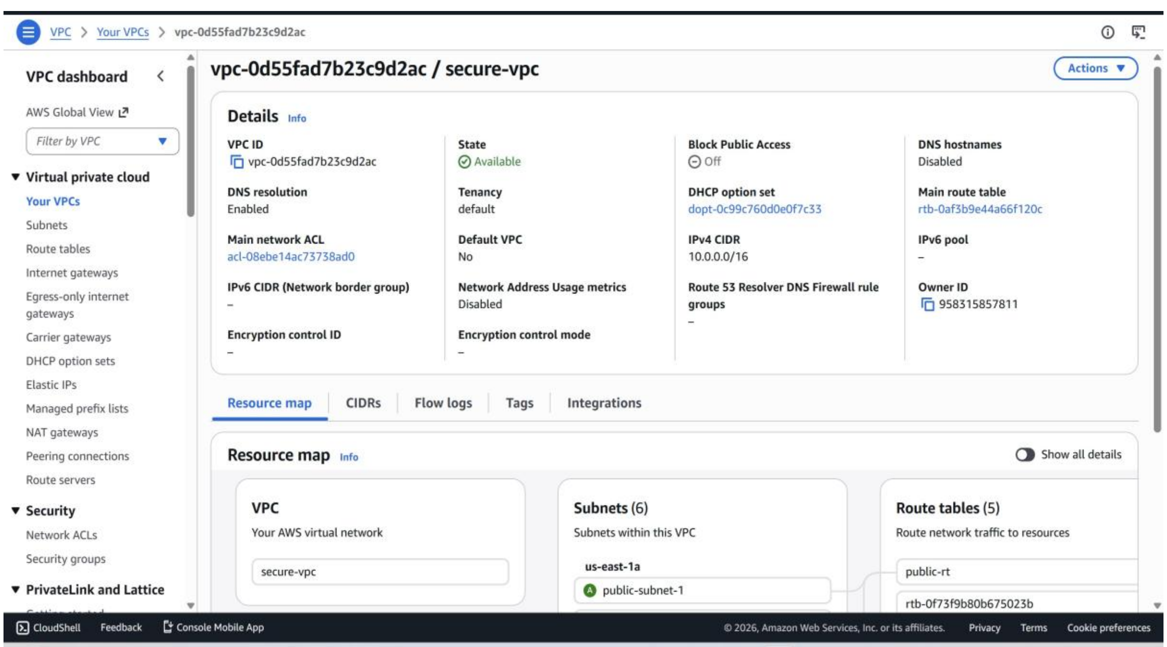
AWS SECURE CLOUD ARCHITECTURE

Evidence Screenshots — Infrastructure Verification

These screenshots provide visual verification of the AWS infrastructure built for the Defense-in-Depth security architecture project. Each screenshot corresponds to a specific component documented in the main architecture document.

1. VPC Configuration

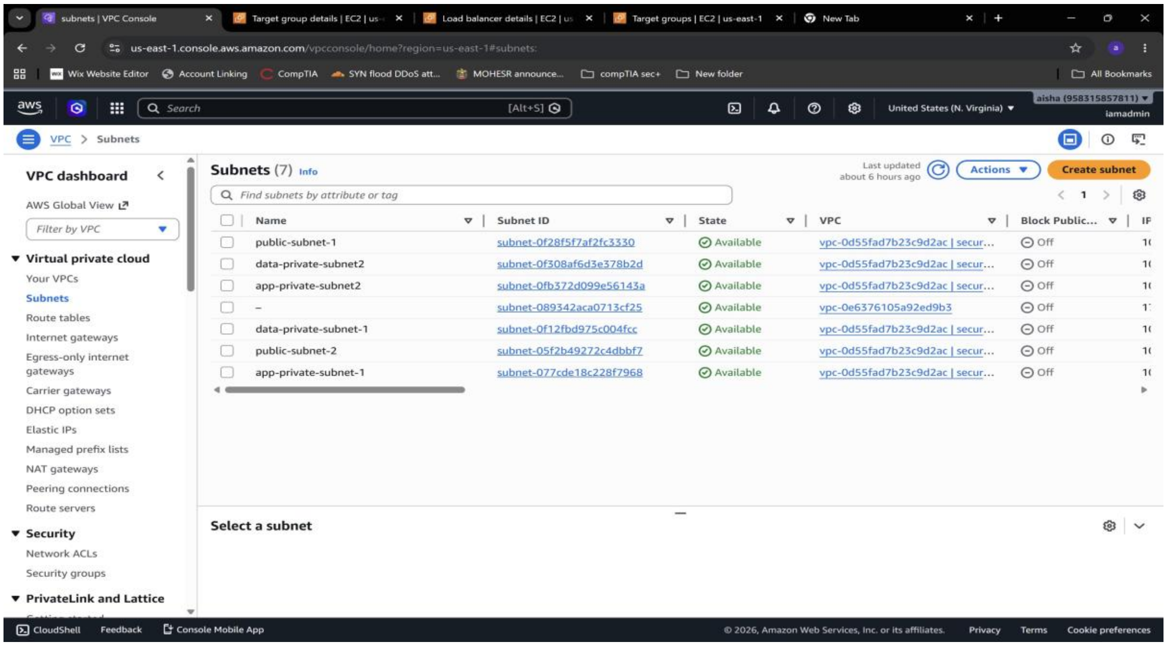
The secure-vpc was created with CIDR 10.0.0.0/16, providing an isolated network environment for all architecture components.



Screenshot: secure-vpc — VPC ID vpc-0d55fad7b23c9d2ac, CIDR 10.0.0.0/16

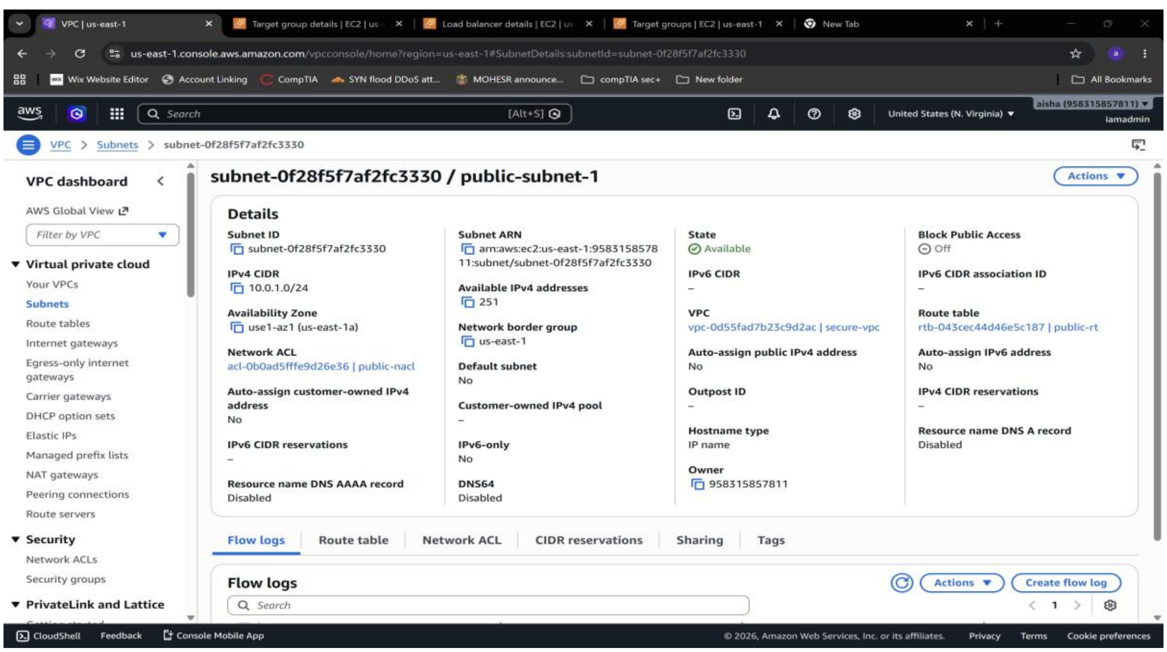
2. Subnet Configuration

Six subnets were created across two Availability Zones — two public, two application private, and two data private — implementing the 3-tier network segmentation.



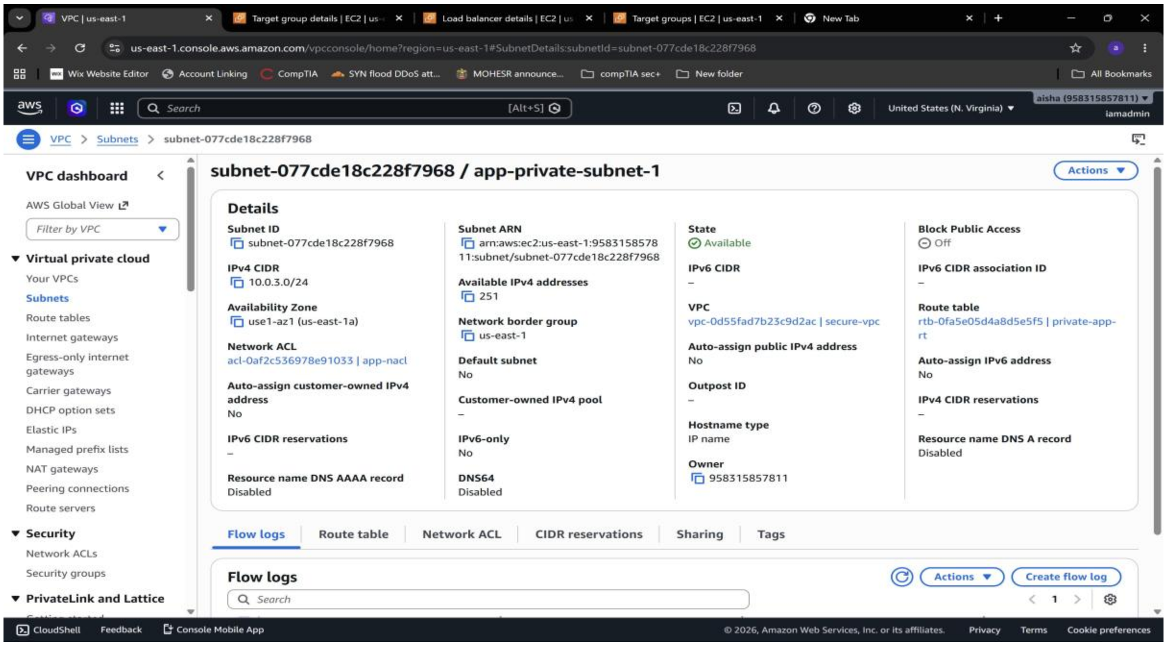
Screenshot: All 6 subnets — public, app-private, and data-private tiers across AZ-a and AZ-b

2.1 Public Subnet



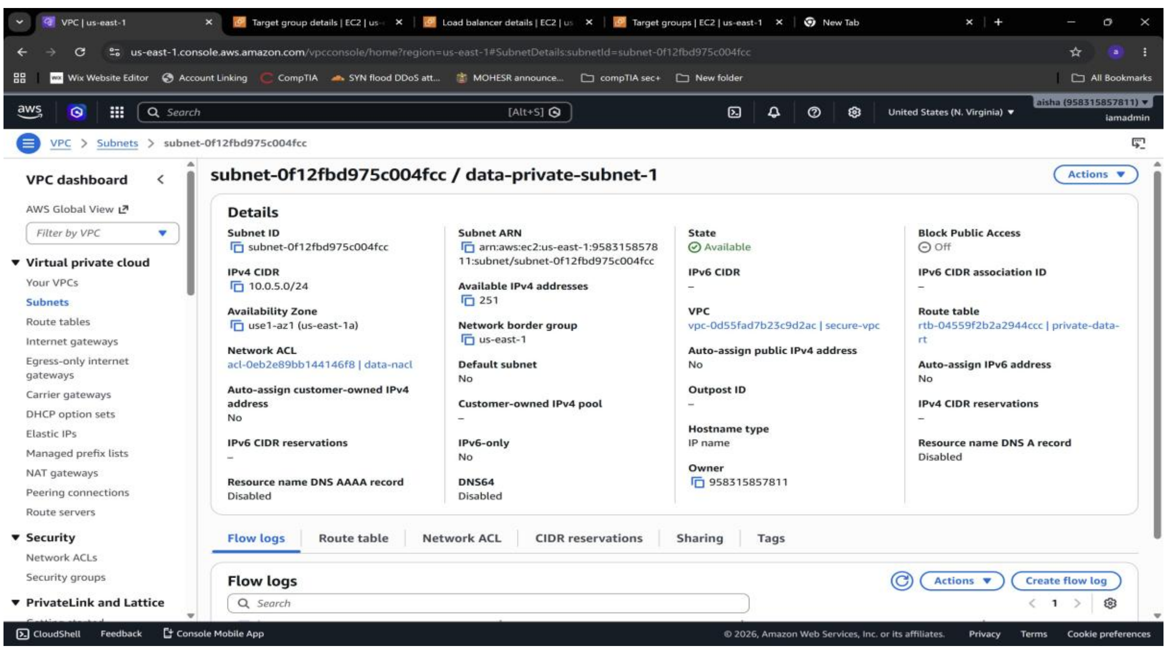
Screenshot: public-subnet-1 — CIDR 10.0.1.0/24, AZ us-east-1a, associated with public-rt

2.2 App Private Subnet



Screenshot: app-private-subnet-1 — CIDR 10.0.3.0/24, AZ us-east-1a, no public IP assignment

2.3 Data Private Subnet

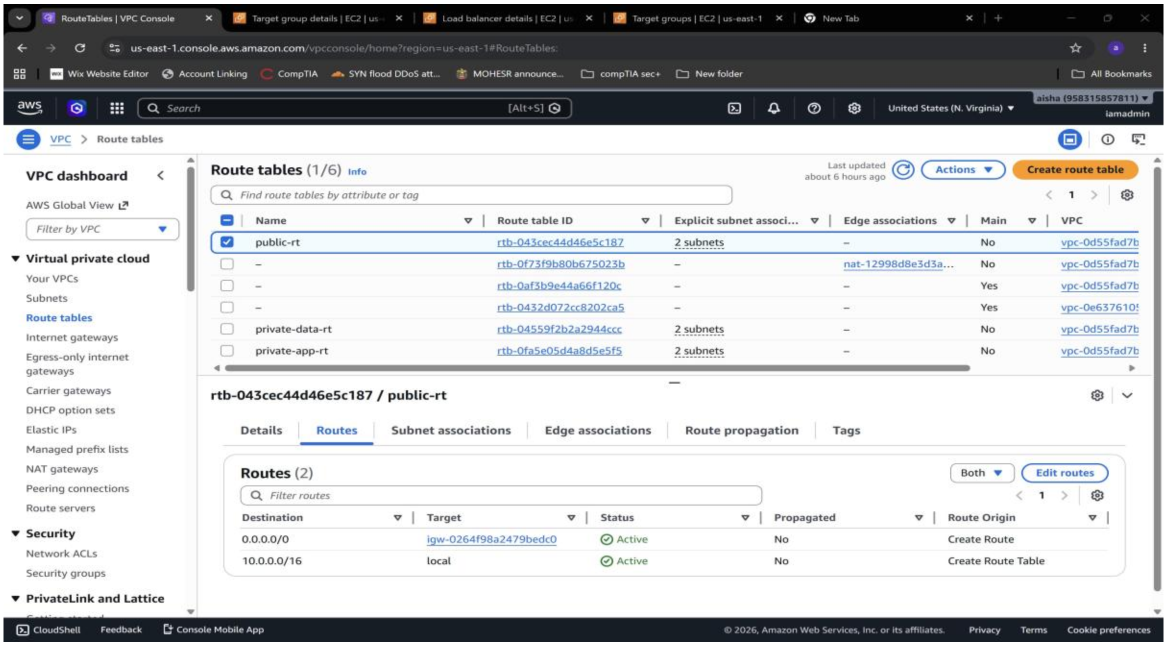


Screenshot: data-private-subnet-1 — CIDR 10.0.5.0/24, AZ us-east-1a, associated with data-nacl

3. Route Tables

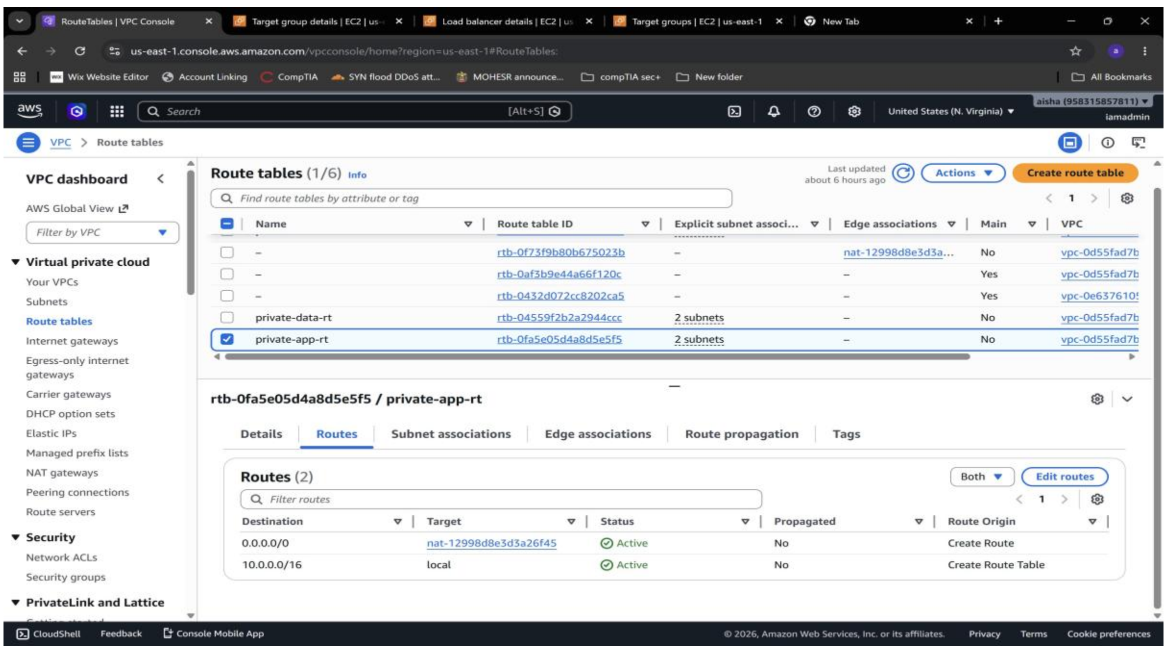
Three route tables implement tiered internet access — public subnets route to IGW, app subnets route to NAT Gateway, data subnets have local-only routing.

3.1 Public Route Table



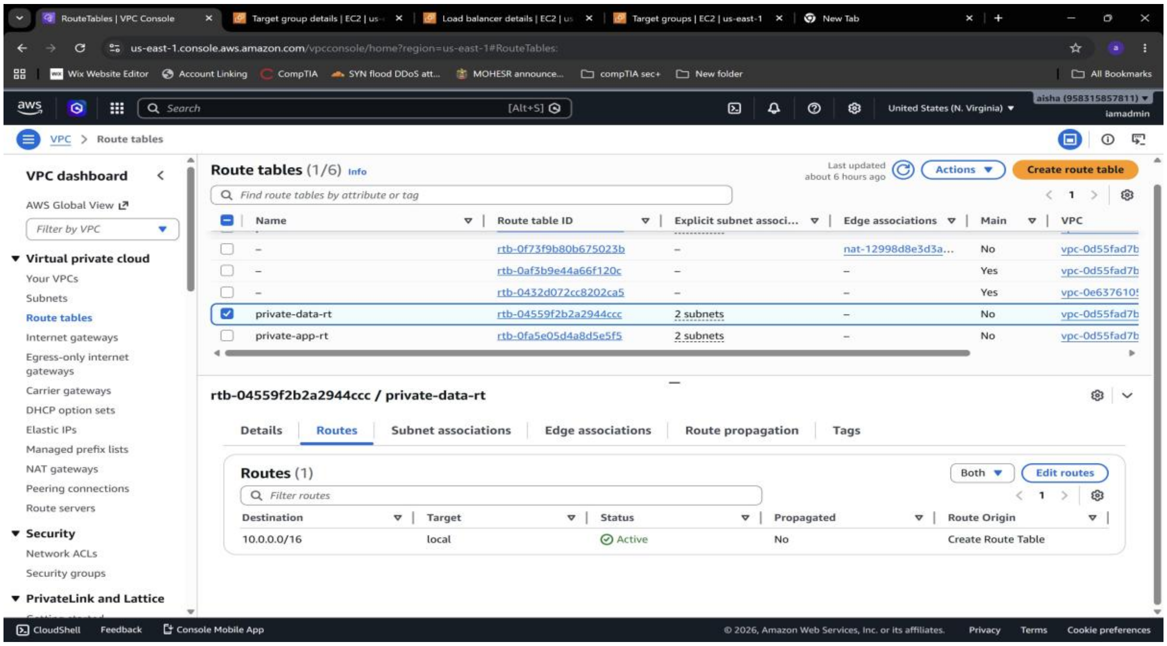
Screenshot: public-rt — routes: 0.0.0.0/0 → IGW, 10.0.0.0/16 → local

3.2 App Private Route Table



Screenshot: private-app-rt — routes: 0.0.0.0/0 → NAT Gateway, 10.0.0.0/16 → local

3.3 Data Private Route Table

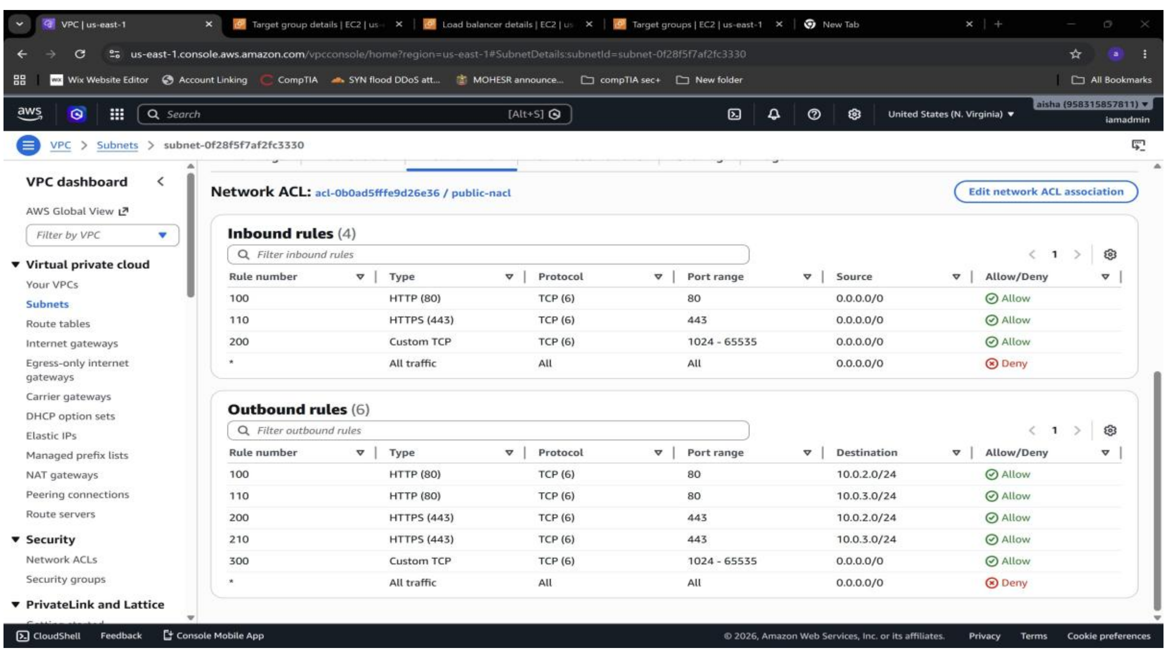


Screenshot: private-data-rt — routes: 10.0.0.0/16 → local only (no internet access)

4. Network Access Control Lists (NACLs)

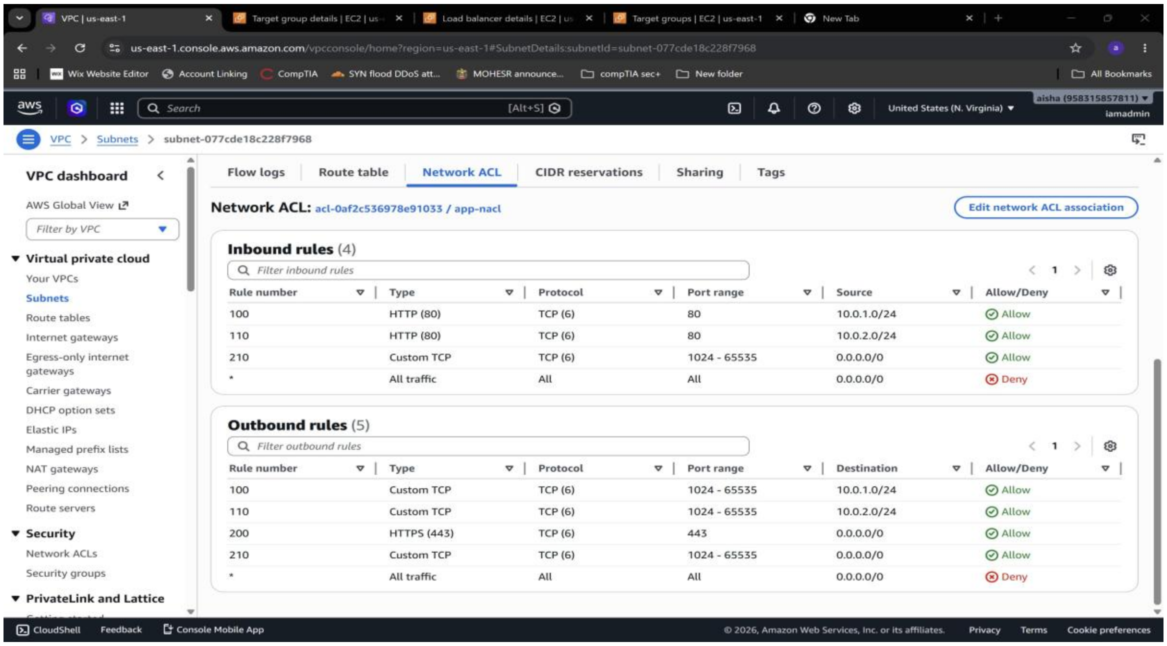
Three NACLs provide stateless subnet-level packet filtering. Each NACL is scoped to allow only the minimum required traffic for its tier.

4.1 Public NACL



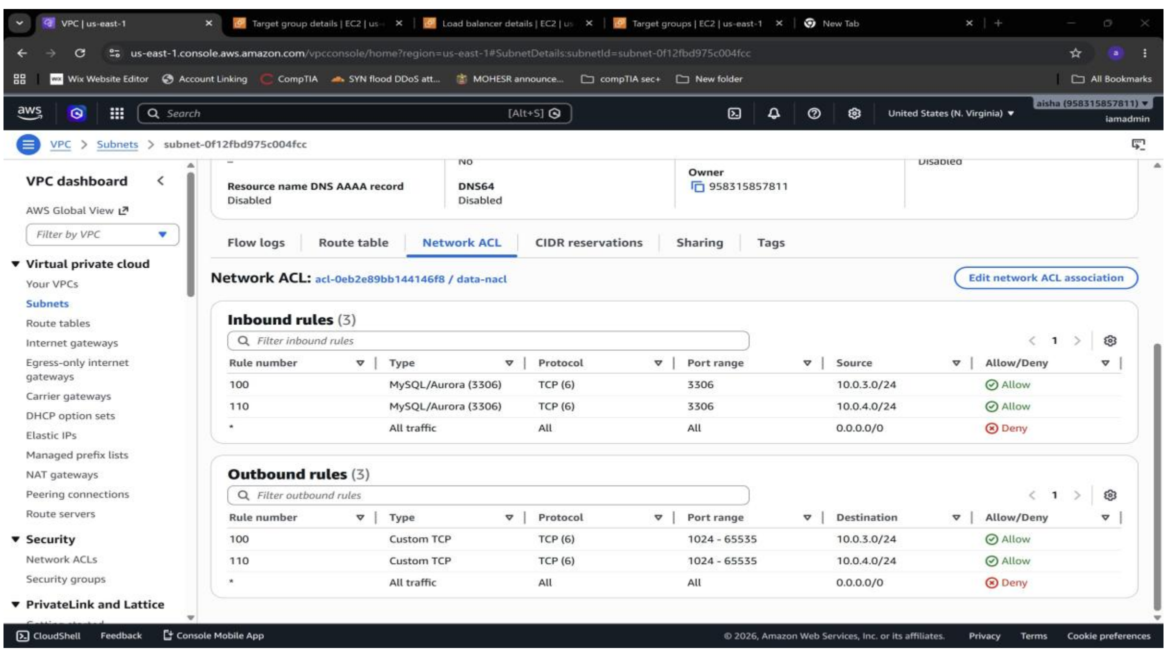
Screenshot: public-nacl — inbound: HTTP/HTTPS/ephemeral from internet; outbound: to app subnets and internet

4.2 App NACL



Screenshot: app-nacl — inbound: port 80 from public subnets only; outbound: responses + HTTPS for SSM/updates

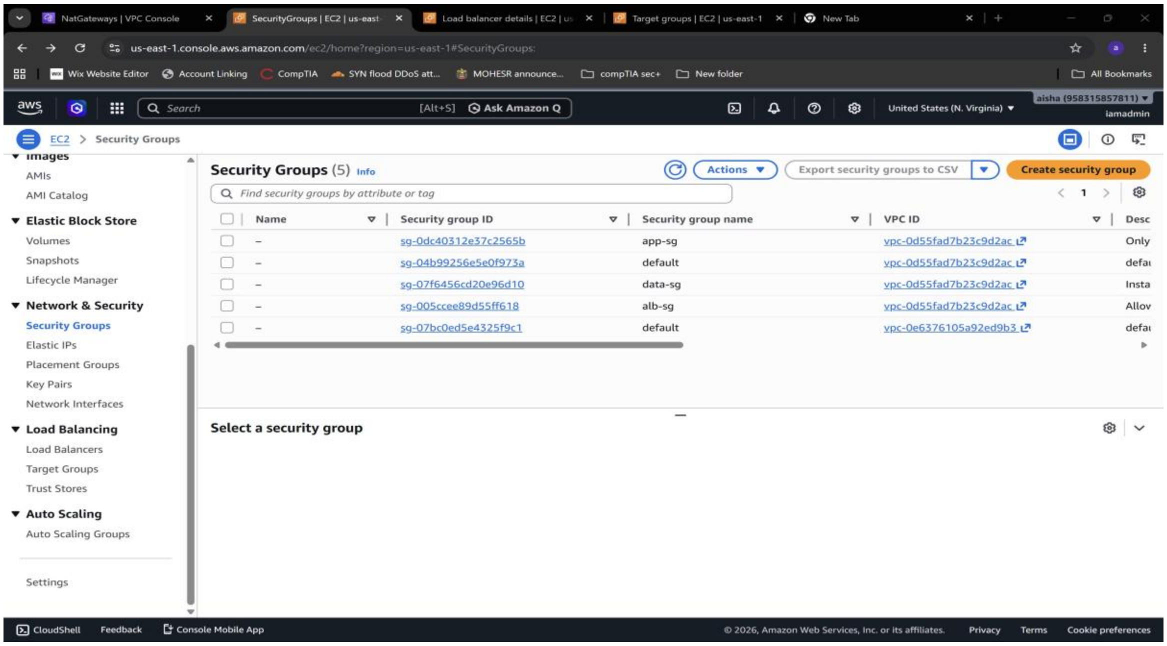
4.3 Data NACL



Screenshot: data-nacl — inbound: port 3306 from app subnets only; outbound: ephemeral ports to app subnets only

5. Security Groups

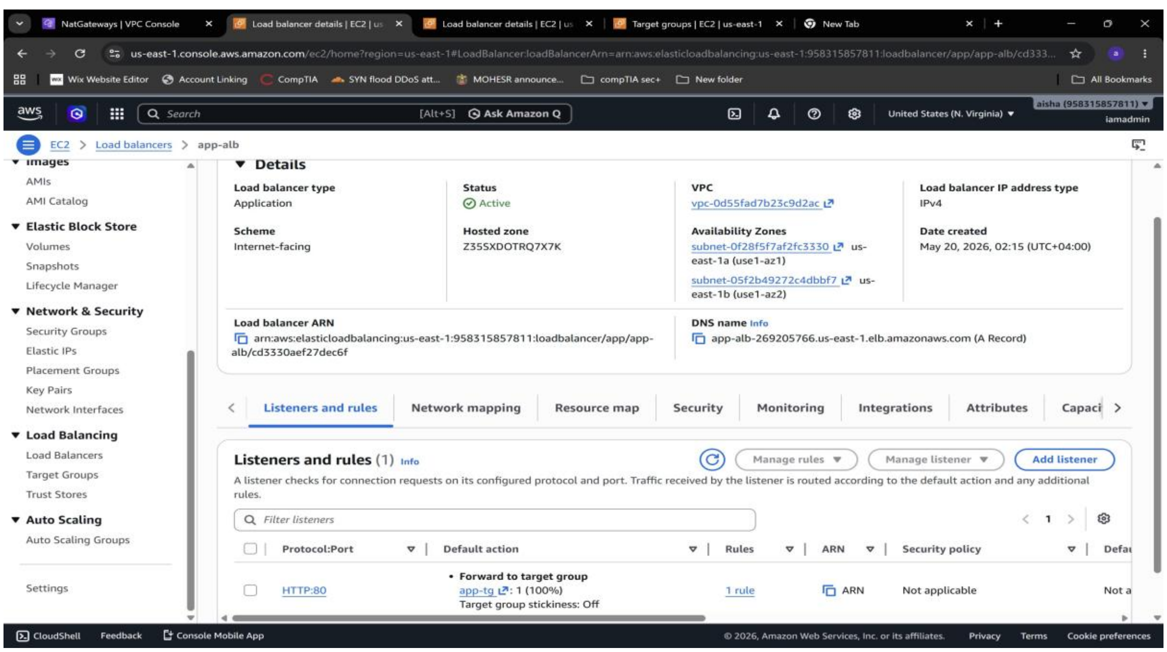
Four security groups implement stateful instance-level access control with SG-to-SG referencing — no IP-based rules used anywhere.



Screenshot: Security Groups — alb-sg, app-sg, data-sg all in secure-vpc

6. Application Load Balancer

The Application Load Balancer is deployed in public subnets across two AZs, configured as internet-facing with HTTP:80 listener forwarding to the app-tg target group.



Screenshot: app-alb — Internet-facing ALB, Active status, deployed across us-east-1a and us-east-1b